

HP1 Analysis: FM action_weight=1 vs Baselines

Three-way comparison — A: Diffusion S6 · B: FM S6 (aw=10) · C: FM S6 (aw=1, HP1)

A = Diffusion · Seed 6 ·
GaussianDiffusion/6

B = FM (aw=10) · Seed 6 · original
hyperparameter

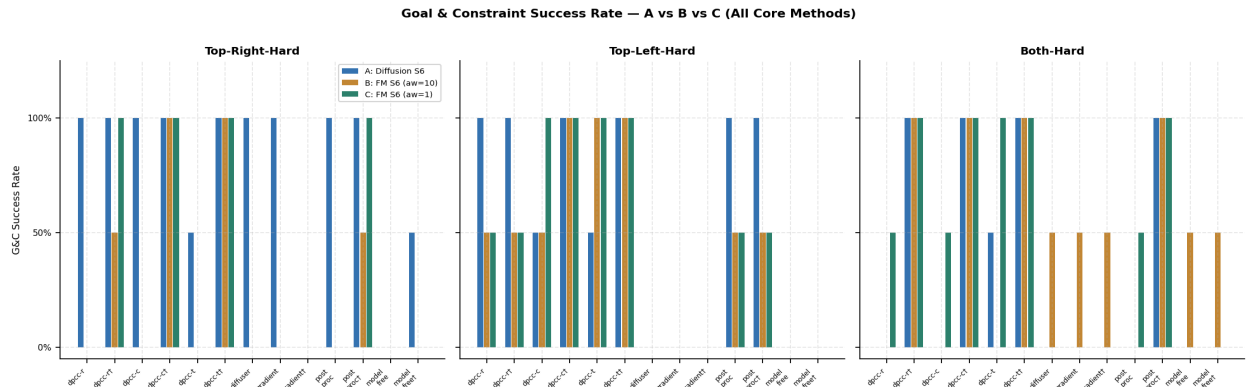
C = FM (aw=1) · Seed 6 · HP1 tune
(this file)

C uses the FM model with **action_weight=1** (changed from 10 in B), all other settings identical. Seed 6, same 3 scenarios, same 17 methods. The key question: does reducing action_weight improve constraint satisfaction and goal success?

Section 1 — Visual Analysis: A vs B vs C

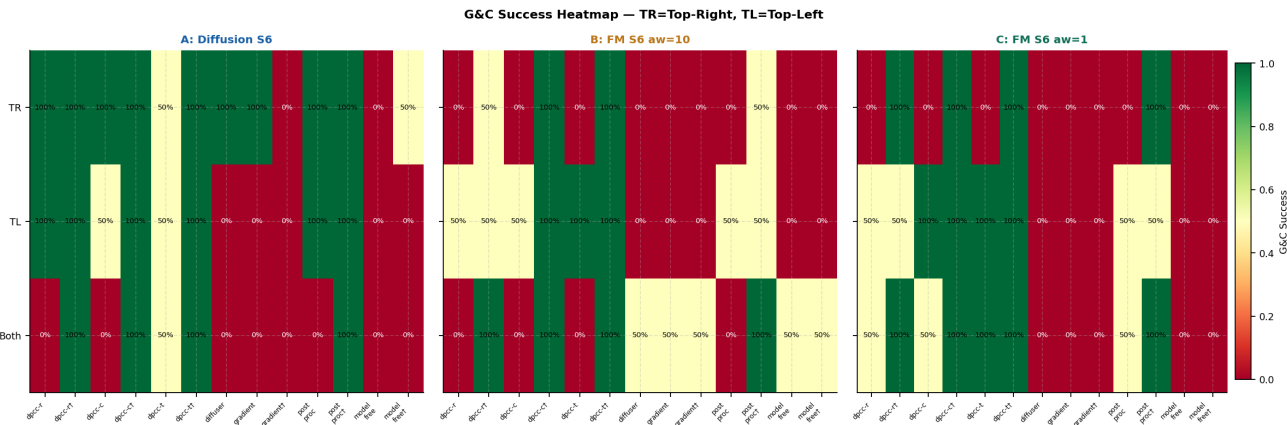
1.1 Goal & Constraint Success Rate — All Core Methods

Primary metric. Blue=A (Diffusion), orange=B (FM aw=10), teal=C (FM aw=1). †=tightened.



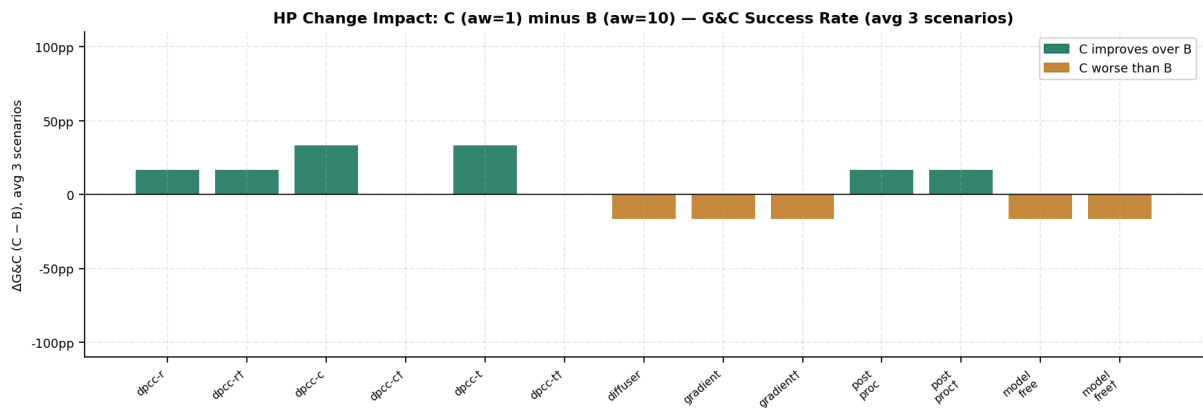
1.2 Heatmap — G&C; Success Rate: A vs B vs C

Each cell = G&C;%. Green=100%, yellow=50%, red=0%. TR=Top-Right, TL=Top-Left.



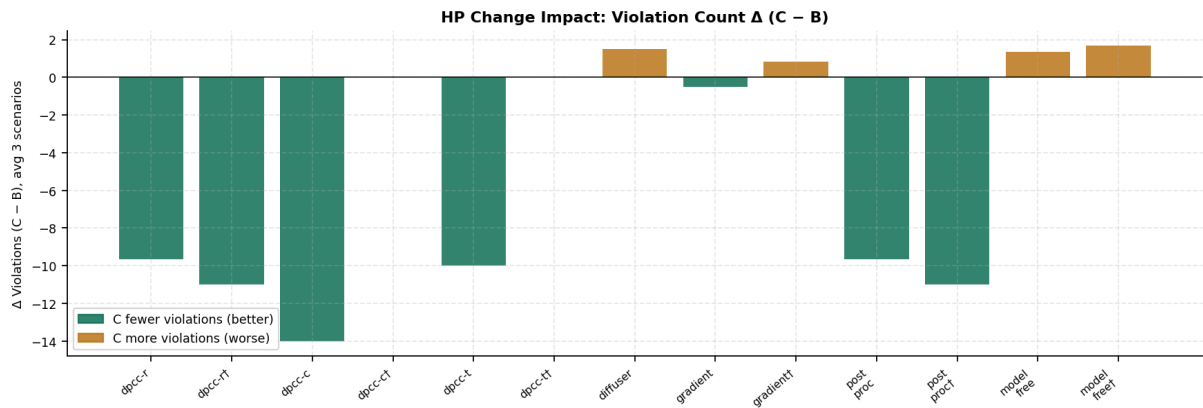
1.3 HP Change Impact: C minus B — G&C; Success Rate

Teal bars = C improves over B (aw=1 better). Orange bars = C worse than B. Averaged over 3 scenarios.



1.4 HP Change Impact: Constraint Violations $\Delta(C - B)$

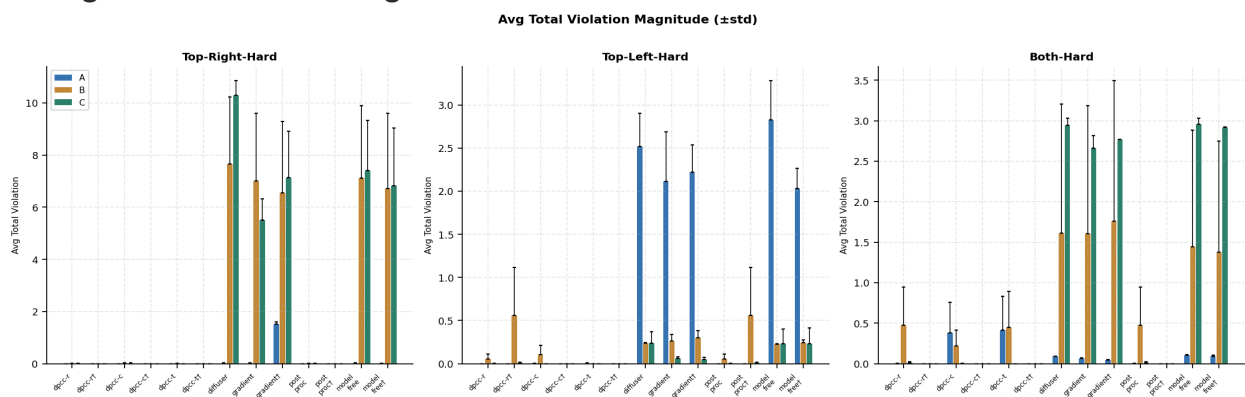
Teal bars = C has fewer violations (better). Averaged over 3 scenarios.



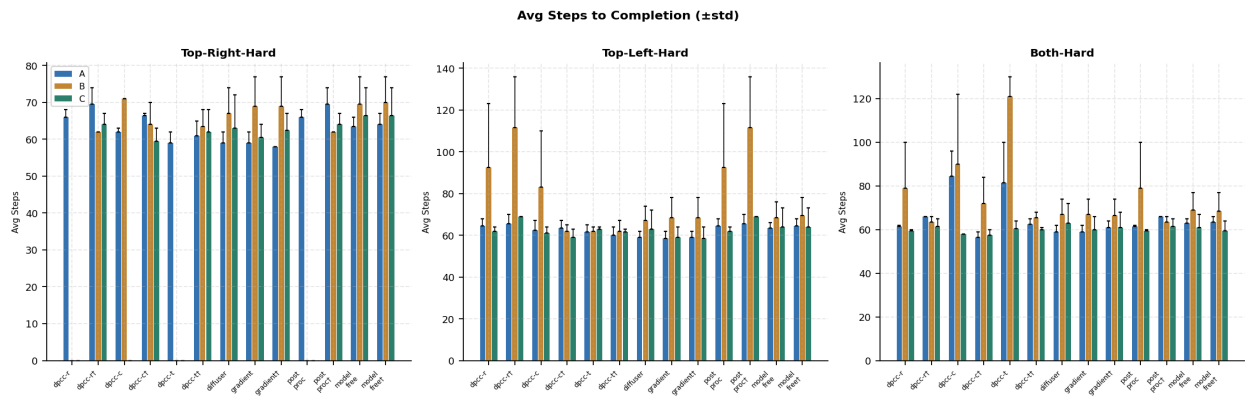
Error bars = $\pm$ std across seeds.



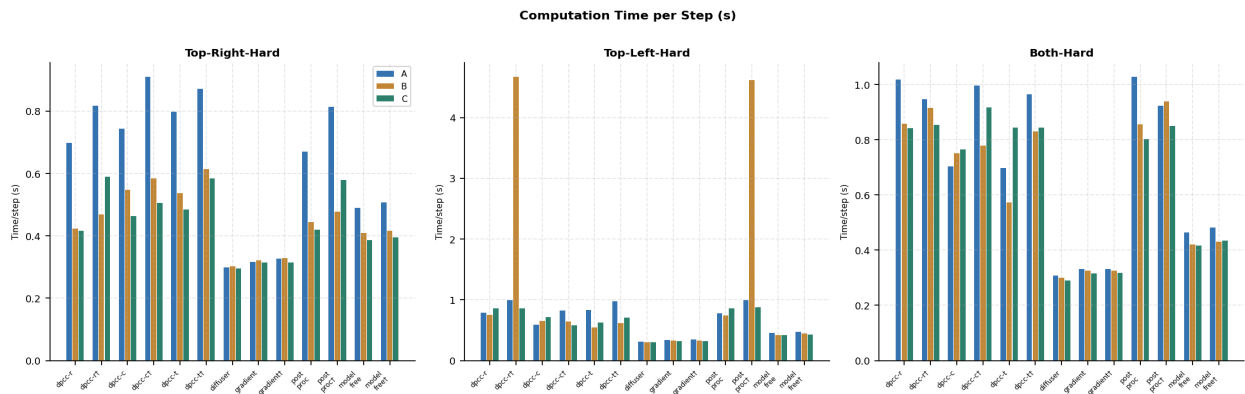
1.6 Avg Total Violation Magnitude



1.7 Average Steps to Completion ($\pm$ std)

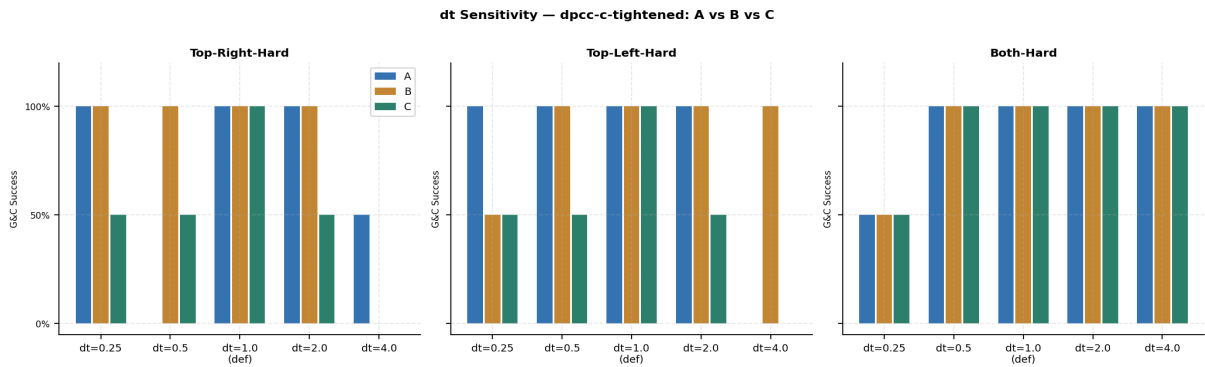


1.8 Computation Time per Step



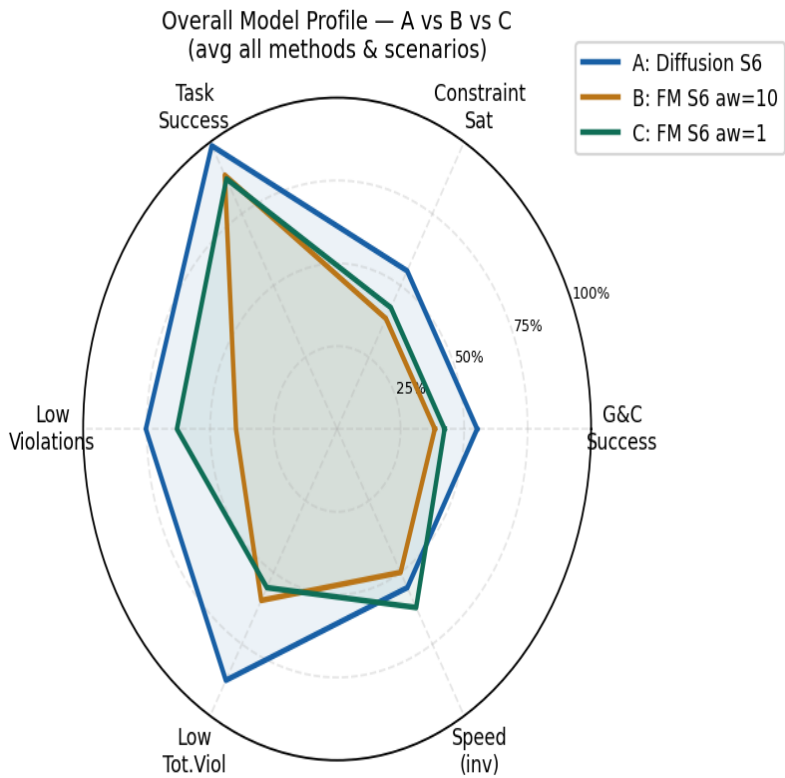
1.9 dt Sensitivity — dpcc-c-tightened: A vs B vs C

G&C; success across $dt \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$.



1.10 Overall Model Profile Radar

Aggregated across all core methods and scenarios.



Section 2 — Complete Raw Data Tables

All metrics from the HP1 log (C), alongside A and B for reference. **SR**=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps±std**, **Viol#±std**, **ViolΣ±std**, **Time** (s/step), **TE**=Tracking Error.

2.1 Top-Right-Hard

C: FM aw=1 — Seed 6 (HP1)

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	0.5±0.5	0.006±0.006	0.417	–
dpcc-r-tight	100%	100%	100%	64.0±3.0	0.0±0.0	0.000±0.000	0.590	–
dpcc-c	0%	0%	0%	0.0±0.0	1.0±1.0	0.015±0.015	0.464	–
dpcc-c-tight	100%	100%	100%	59.5±3.5	0.0±0.0	0.000±0.000	0.505	–
dpcc-t	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.485	–
dpcc-t-tight	100%	100%	100%	62.0±6.0	0.0±0.0	0.000±0.000	0.585	–
diffuser	100%	0%	0%	63.0±9.0	30.5±1.5	10.286±0.579	0.296	0.032
gradient	100%	0%	0%	60.5±3.5	30.0±4.0	5.513±0.805	0.314	–
gradient-tight	100%	0%	0%	62.5±4.5	32.0±3.0	7.143±1.767	0.315	–
post_proc	0%	0%	0%	0.0±0.0	0.5±0.5	0.006±0.006	0.419	–
post_proc-tight	100%	100%	100%	64.0±3.0	0.0±0.0	0.000±0.000	0.579	–
model_free	100%	0%	0%	66.5±7.5	33.0±3.0	7.406±1.922	0.386	–
model_free-tight	100%	0%	0%	66.5±7.5	33.0±3.0	6.828±2.201	0.395	–
dpcc-c-tight-dt0.25	50%	50%	50%	82.0±0.0	1.0±1.0	0.006±0.004	1.027	–
dpcc-c-tight-dt0.5	50%	50%	50%	62.0±0.0	0.0±0.0	0.001±0.001	0.884	–
dpcc-c-tight-dt2.0	50%	50%	50%	57.0±0.0	0.0±0.0	0.000±0.000	0.479	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.462	–

B: FM aw=10 — Seed 6 (original)

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	0.5±0.5	0.002±0.002	0.424	–
dpcc-r-tight	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.469	–
dpcc-c	50%	0%	0%	71.0±0.0	2.0±2.0	0.015±0.015	0.547	–
dpcc-c-tight	100%	100%	100%	64.0±6.0	0.0±0.0	0.000±0.000	0.585	–
dpcc-t	0%	0%	0%	0.0±0.0	1.0±1.0	0.010±0.010	0.537	–

dpcc-t-tight	100 %	100 %	100 %	63.5±4.5	0.0±0.0	0.000±0.000	0.615	–
diffuser	100 %	0%	0%	67.0±7.0	33.0±3.0	7.654±2.573	0.303	0.035
gradient	100 %	0%	0%	69.0±8.0	34.0±3.0	7.014±2.598	0.321	–
gradient-tight	100 %	0%	0%	69.0±8.0	32.0±2.0	6.556±2.728	0.329	–
post_proc	0%	0%	0%	0.0±0.0	0.5±0.5	0.002±0.002	0.444	–
post_proc-tight	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.478	–
model_free	100 %	0%	0%	69.5±7.5	35.5±2.5	7.118±2.776	0.409	–
model_free-tight	100 %	0%	0%	70.0±7.0	35.0±2.0	6.726±2.879	0.416	–
dpcc-c-tight-dt0.25	100 %	100 %	100 %	82.0±5.0	0.0±0.0	0.003±0.002	1.644	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	74.5±6.5	0.0±0.0	0.001±0.001	1.163	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	69.5±8.5	0.0±0.0	0.000±0.000	0.500	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.491	–

A: Diffusion — Seed 6

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100 %	100 %	100 %	66.0±2.0	0.0±0.0	0.000±0.000	0.698	–
dpcc-r-tight	100 %	100 %	100 %	69.5±4.5	0.0±0.0	0.000±0.000	0.817	–
dpcc-c	100 %	100 %	100 %	62.0±1.0	0.0±0.0	0.000±0.000	0.744	–
dpcc-c-tight	100 %	100 %	100 %	66.5±0.5	0.0±0.0	0.000±0.000	0.911	–
dpcc-t	100 %	50%	50%	59.0±3.0	0.5±0.5	0.000±0.000	0.799	–
dpcc-t-tight	100 %	100 %	100 %	61.0±4.0	0.0±0.0	0.000±0.000	0.871	–
diffuser	100 %	100 %	100 %	59.0±3.0	0.0±0.0	0.021±0.002	0.298	0.025
gradient	100 %	100 %	100 %	59.0±3.0	0.0±0.0	0.021±0.002	0.316	–
gradient-tight	50%	0%	0%	58.0±0.0	15.0±0.0	1.521±0.082	0.327	–
post_proc	100 %	100 %	100 %	66.0±2.0	0.0±0.0	0.000±0.000	0.670	–
post_proc-tight	100 %	100 %	100 %	69.5±4.5	0.0±0.0	0.000±0.000	0.814	–

model_free	100 %	0%	0%	63.5±2.5	3.0±1.0	0.012±0.011	0.490	–
model_free-tight	100 %	50%	50%	64.0±3.0	2.0±2.0	0.009±0.009	0.508	–
dpcc-c-tight-dt0.25	100 %	100 %	100 %	81.0±6.0	0.0±0.0	0.002±0.000	2.225	–
dpcc-c-tight-dt0.5	100 %	0%	0%	75.5±2.5	1.0±0.0	0.003±0.000	1.668	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	66.5±0.5	0.0±0.0	0.000±0.000	0.689	–
dpcc-c-tight-dt4.0	50%	50%	50%	72.0±0.0	0.0±0.0	0.000±0.000	0.535	–

2.2 Top-Left-Hard

C: FM aw=1 — Seed 6 (HP1)

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100 %	50%	50%	62.0±2.0	0.5±0.5	0.001±0.001	0.856	–
dpcc-r-tight	50%	50%	50%	69.0±0.0	1.0±1.0	0.007±0.007	0.859	–
dpcc-c	100 %	100 %	100 %	61.0±3.0	0.0±0.0	0.000±0.000	0.711	–
dpcc-c-tight	100 %	100 %	100 %	59.0±4.0	0.0±0.0	0.000±0.000	0.577	–
dpcc-t	100 %	100 %	100 %	63.0±1.0	0.0±0.0	0.000±0.000	0.622	–
dpcc-t-tight	100 %	100 %	100 %	61.5±1.5	0.0±0.0	0.000±0.000	0.704	–
diffuser	100 %	0%	0%	63.0±9.0	11.5±2.5	0.235±0.137	0.303	0.032
gradient	100 %	0%	0%	59.0±5.0	8.0±1.0	0.063±0.015	0.318	–
gradient-tight	100 %	0%	0%	58.5±5.5	5.5±2.5	0.048±0.027	0.316	–
post_proc	100 %	50%	50%	62.0±2.0	0.5±0.5	0.001±0.001	0.862	–
post_proc-tight	50%	50%	50%	69.0±0.0	1.0±1.0	0.007±0.007	0.874	–
model_free	100 %	0%	0%	64.0±9.0	11.0±3.0	0.232±0.170	0.415	–
model_free-tight	100 %	0%	0%	64.0±9.0	11.0±3.0	0.233±0.178	0.425	–
dpcc-c-tight-dt0.25	50%	50%	50%	70.0±0.0	0.0±0.0	0.002±0.001	1.147	–
dpcc-c-tight-dt0.5	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.811	–
dpcc-c-tight-dt2.0	50%	50%	50%	58.0±0.0	0.0±0.0	0.000±0.000	0.478	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.413	–

B: FM aw=10 — Seed 6 (original)

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100 %	50%	50%	92.5±30.5	7.0±7.0	0.055±0.055	0.753	–
dpcc-r-tight	100 %	50%	50%	111.5±24.5	34.0±34.0	0.557±0.557	4.675	–
dpcc-c	100 %	50%	50%	83.0±27.0	15.0±15.0	0.105±0.105	0.656	–
dpcc-c-tight	100 %	100 %	100 %	62.0±3.0	0.0±0.0	0.000±0.000	0.646	–
dpcc-t	100 %	100 %	100 %	62.0±2.0	0.0±0.0	0.000±0.000	0.548	–
dpcc-t-tight	100 %	100 %	100 %	62.0±5.0	0.0±0.0	0.000±0.000	0.614	–
diffuser	100 %	0%	0%	67.0±7.0	11.5±0.5	0.237±0.005	0.304	0.035
gradient	100 %	0%	0%	68.5±9.5	12.0±1.0	0.263±0.077	0.326	–
gradient-tight	100 %	0%	0%	68.5±9.5	12.5±1.5	0.299±0.085	0.324	–
post_proc	100 %	50%	50%	92.5±30.5	7.0±7.0	0.055±0.055	0.744	–
post_proc-tight	100 %	50%	50%	111.5±24.5	34.0±34.0	0.557±0.557	4.615	–
model_free	100 %	0%	0%	68.5±7.5	11.5±0.5	0.225±0.008	0.422	–
model_free-tight	100 %	0%	0%	69.5±8.5	12.0±1.0	0.245±0.028	0.441	–
dpcc-c-tight-dt0.25	100 %	50%	50%	89.0±7.0	1.0±1.0	0.013±0.009	1.071	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	66.5±0.5	0.0±0.0	0.000±0.000	0.563	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	68.0±8.0	0.0±0.0	0.000±0.000	0.563	–
dpcc-c-tight-dt4.0	100 %	100 %	100 %	120.0±6.0	0.0±0.0	0.000±0.000	0.486	–

A: Diffusion — Seed 6

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100 %	100 %	100 %	64.5±3.5	0.0±0.0	0.000±0.000	0.784	–
dpcc-r-tight	100 %	100 %	100 %	65.5±4.5	0.0±0.0	0.000±0.000	0.998	–
dpcc-c	100 %	50%	50%	62.5±4.5	1.0±1.0	0.003±0.003	0.592	–
dpcc-c-tight	100 %	100 %	100 %	63.5±3.5	0.0±0.0	0.000±0.000	0.822	–

dpcc-t	100 %	50%	50%	61.5±3.5	1.5±1.5	0.004±0.004	0.829	–
dpcc-t-tight	100 %	100 %	100 %	60.0±4.0	0.0±0.0	0.000±0.000	0.976	–
diffuser	100 %	0%	0%	59.0±3.0	20.0±2.0	2.515±0.387	0.306	0.025
gradient	100 %	0%	0%	58.5±3.5	17.5±3.5	2.116±0.571	0.334	–
gradient-tight	100 %	0%	0%	59.0±3.0	19.5±1.5	2.218±0.316	0.348	–
post_proc	100 %	100 %	100 %	64.5±3.5	0.0±0.0	0.000±0.000	0.782	–
post_proc-tight	100 %	100 %	100 %	65.5±4.5	0.0±0.0	0.000±0.000	0.996	–
model_free	100 %	0%	0%	63.5±2.5	19.0±0.0	2.829±0.455	0.458	–
model_free-tight	100 %	0%	0%	64.5±3.5	15.0±2.0	2.029±0.233	0.470	–
dpcc-c-tight-dt0.25	100 %	100 %	100 %	70.0±3.0	0.0±0.0	0.000±0.000	1.259	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	63.5±0.5	0.0±0.0	0.000±0.000	0.937	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	64.5±0.5	0.0±0.0	0.000±0.000	0.679	–
dpcc-c-tight-dt4.0	100 %	0%	0%	107.0±2.0	16.5±0.5	3.711±0.053	0.810	–

2.3 Both-Hard

C: FM aw=1 — Seed 6 (HP1)

Method	SR	CS	G&C ;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100 %	50%	50%	59.5±0.5	2.0±2.0	0.011±0.011	0.842	–
dpcc-r-tight	100 %	100 %	100 %	61.5±3.5	0.0±0.0	0.000±0.000	0.855	–
dpcc-c	100 %	50%	50%	58.0±0.0	0.5±0.5	0.003±0.003	0.766	–
dpcc-c-tight	100 %	100 %	100 %	57.5±2.5	0.0±0.0	0.000±0.000	0.918	–
dpcc-t	100 %	100 %	100 %	60.5±3.5	0.0±0.0	0.000±0.000	0.845	–
dpcc-t-tight	100 %	100 %	100 %	60.0±1.0	0.0±0.0	0.000±0.000	0.845	–
diffuser	100 %	0%	0%	63.0±9.0	15.0±2.0	2.946±0.087	0.289	0.032
gradient	100 %	0%	0%	60.0±6.0	13.5±0.5	2.661±0.152	0.315	–

gradient-tight	100 %	0%	0%	61.0±7.0	16.5±3.5	2.765±0.002	0.318	–
post_proc	100 %	50%	50%	59.5±0.5	2.0±2.0	0.011±0.011	0.803	–
post_proc-tight	100 %	100 %	100 %	61.5±3.5	0.0±0.0	0.000±0.000	0.850	–
model_free	100 %	0%	0%	61.0±6.0	14.0±1.0	2.955±0.076	0.417	–
model_free-tight	100 %	0%	0%	59.5±4.5	14.0±1.0	2.916±0.004	0.435	–
dpcc-c-tight-dt0.25	100 %	50%	50%	75.5±1.5	1.0±1.0	0.014±0.012	1.560	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	62.5±4.5	0.0±0.0	0.001±0.000	1.133	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	62.0±3.0	0.0±0.0	0.000±0.000	0.739	–
dpcc-c-tight-dt4.0	100 %	100 %	100 %	79.5±5.5	0.0±0.0	0.000±0.000	0.601	–

B: FM aw=10 — Seed 6 (original)

Method	SR	CS	G&C ;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100 %	0%	0%	79.0±21.0	24.5±22.5	0.473±0.469	0.859	–
dpcc-r-tight	100 %	100 %	100 %	63.5±2.5	0.0±0.0	0.000±0.000	0.916	–
dpcc-c	100 %	0%	0%	90.0±32.0	26.5±21.5	0.217±0.197	0.751	–
dpcc-c-tight	100 %	100 %	100 %	72.0±12.0	0.0±0.0	0.000±0.000	0.779	–
dpcc-t	100 %	0%	0%	121.0±9.0	29.0±27.0	0.445±0.444	0.573	–
dpcc-t-tight	100 %	100 %	100 %	65.5±2.5	0.0±0.0	0.000±0.000	0.830	–
diffuser	100 %	50%	50%	67.0±7.0	8.0±8.0	1.612±1.594	0.300	0.035
gradient	100 %	50%	50%	67.0±7.0	7.0±7.0	1.602±1.584	0.326	–
gradient-tight	100 %	50%	50%	66.5±7.5	7.0±7.0	1.757±1.740	0.325	–
post_proc	100 %	0%	0%	79.0±21.0	24.5±22.5	0.473±0.469	0.857	–
post_proc-tight	100 %	100 %	100 %	63.5±2.5	0.0±0.0	0.000±0.000	0.940	–
model_free	100 %	50%	50%	69.0±8.0	7.0±7.0	1.441±1.441	0.421	–
model_free-tight	100 %	50%	50%	68.5±8.5	6.0±6.0	1.375±1.375	0.431	–

dpcc-c-tight-dt0.25	100 %	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–
dpcc-c-tight-dt4.0	100 %	100 %	100 %	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–

A: Diffusion — Seed 6

Method	SR	CS	G&C ;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	100 %	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.019	–
dpcc-r-tight	100 %	100 %	100 %	66.0±0.0	0.0±0.0	0.000±0.000	0.948	–
dpcc-c	100 %	0%	0%	84.5±11.5	19.0±17.0	0.381±0.377	0.704	–
dpcc-c-tight	100 %	100 %	100 %	56.5±2.5	0.0±0.0	0.000±0.000	0.997	–
dpcc-t	100 %	50%	50%	81.5±18.5	17.0±17.0	0.416±0.416	0.697	–
dpcc-t-tight	100 %	100 %	100 %	62.5±2.5	0.0±0.0	0.000±0.000	0.964	–
diffuser	100 %	0%	0%	59.0±3.0	7.5±0.5	0.090±0.001	0.307	0.025
gradient	100 %	0%	0%	59.0±3.0	7.0±1.0	0.064±0.007	0.331	–
gradient-tight	100 %	0%	0%	61.0±3.0	6.0±1.0	0.044±0.004	0.332	–
post_proc	100 %	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.029	–
post_proc-tight	100 %	100 %	100 %	66.0±0.0	0.0±0.0	0.000±0.000	0.924	–
model_free	100 %	0%	0%	63.0±2.0	8.5±0.5	0.103±0.008	0.465	–
model_free-tight	100 %	0%	0%	63.5±2.5	8.5±0.5	0.091±0.010	0.481	–
dpcc-c-tight-dt0.25	100 %	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100 %	100 %	100 %	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100 %	100 %	100 %	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–
dpcc-c-tight-dt4.0	100 %	100 %	100 %	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–

Section 3 — Three-Way Head-to-Head Comparison

For every method × scenario: G&C; and violation counts for A, B, and C. Best G&C; highlighted. Green violation cell = lowest count among the three.

Top-Right-Hard

Method	A G&C; C;	B G&C; C;	C G&C; C;	Best	A Viol	B Viol	C Viol	A Steps	B Steps	C Steps
dpcc-r	100 %	0%	0%	A	0.0	0.5	0.5	66.0	0.0	0.0
dpcc-r-tight	100 %	50%	100 %	A/C	0.0	0.0	0.0	69.5	62.0	64.0
dpcc-c	100 %	0%	0%	A	0.0	2.0	1.0	62.0	71.0	0.0
dpcc-c-tight	100 %	100 %	100 %	A/B/ C	0.0	0.0	0.0	66.5	64.0	59.5
dpcc-t	50%	0%	0%	A	0.5	1.0	0.0	59.0	0.0	0.0
dpcc-t-tight	100 %	100 %	100 %	A/B/ C	0.0	0.0	0.0	61.0	63.5	62.0
diffuser	100 %	0%	0%	A	0.0	33.0	30.5	59.0	67.0	63.0
gradient	100 %	0%	0%	A	0.0	34.0	30.0	59.0	69.0	60.5
gradient-tight	0%	0%	0%	A/B/ C	15.0	32.0	32.0	58.0	69.0	62.5
post_proc	100 %	0%	0%	A	0.0	0.5	0.5	66.0	0.0	0.0
post_proc-tight	100 %	50%	100 %	A/C	0.0	0.0	0.0	69.5	62.0	64.0
model_free	0%	0%	0%	A/B/ C	3.0	35.5	33.0	63.5	69.5	66.5
model_free-tight	50%	0%	0%	A	2.0	35.0	33.0	64.0	70.0	66.5
dpcc-c-tight-dt0.25	100 %	100 %	50%	A/B	0.0	0.0	1.0	81.0	82.0	82.0
dpcc-c-tight-dt0.5	0%	100 %	50%	B	1.0	0.0	0.0	75.5	74.5	62.0
dpcc-c-tight-dt2.0	100 %	100 %	50%	A/B	0.0	0.0	0.0	66.5	69.5	57.0
dpcc-c-tight-dt4.0	50%	0%	0%	A	0.0	0.0	0.0	72.0	0.0	0.0

Top-Left-Hard

Method	A G&C; C;	B G&C; C;	C G&C; C;	Best	A Viol	B Viol	C Viol	A Steps	B Steps	C Steps
dpcc-r	100 %	50%	50%	A	0.0	7.0	0.5	64.5	92.5	62.0
dpcc-r-tight	100 %	50%	50%	A	0.0	34.0	1.0	65.5	111.5	69.0

dpcc-c	50%	50%	100%	C	1.0	15.0	0.0	62.5	83.0	61.0
dpcc-c-tight	100%	100%	100%	A/B/C	0.0	0.0	0.0	63.5	62.0	59.0
dpcc-t	50%	100%	100%	B/C	1.5	0.0	0.0	61.5	62.0	63.0
dpcc-t-tight	100%	100%	100%	A/B/C	0.0	0.0	0.0	60.0	62.0	61.5
diffuser	0%	0%	0%	A/B/C	20.0	11.5	11.5	59.0	67.0	63.0
gradient	0%	0%	0%	A/B/C	17.5	12.0	8.0	58.5	68.5	59.0
gradient-tight	0%	0%	0%	A/B/C	19.5	12.5	5.5	59.0	68.5	58.5
post_proc	100%	50%	50%	A	0.0	7.0	0.5	64.5	92.5	62.0
post_proc-tight	100%	50%	50%	A	0.0	34.0	1.0	65.5	111.5	69.0
model_free	0%	0%	0%	A/B/C	19.0	11.5	11.0	63.5	68.5	64.0
model_free-tight	0%	0%	0%	A/B/C	15.0	12.0	11.0	64.5	69.5	64.0
dpcc-c-tight-dt0.25	100%	50%	50%	A	0.0	1.0	0.0	70.0	89.0	70.0
dpcc-c-tight-dt0.5	100%	100%	50%	A/B	0.0	0.0	0.0	63.5	66.5	62.0
dpcc-c-tight-dt2.0	100%	100%	50%	A/B	0.0	0.0	0.0	64.5	68.0	58.0
dpcc-c-tight-dt4.0	0%	100%	0%	B	16.5	0.0	0.0	107.0	120.0	0.0

Both-Hard

Method	A G&C;	B G&C;	C G&C;	Best	A Viol	B Viol	C Viol	A Steps	B Steps	C Steps
dpcc-r	0%	0%	50%	C	2.5	24.5	2.0	61.5	79.0	59.5
dpcc-r-tight	100%	100%	100%	A/B/C	0.0	0.0	0.0	66.0	63.5	61.5
dpcc-c	0%	0%	50%	C	19.0	26.5	0.5	84.5	90.0	58.0
dpcc-c-tight	100%	100%	100%	A/B/C	0.0	0.0	0.0	56.5	72.0	57.5
dpcc-t	50%	0%	100%	C	17.0	29.0	0.0	81.5	121.0	60.5
dpcc-t-tight	100%	100%	100%	A/B/C	0.0	0.0	0.0	62.5	65.5	60.0
diffuser	0%	50%	0%	B	7.5	8.0	15.0	59.0	67.0	63.0
gradient	0%	50%	0%	B	7.0	7.0	13.5	59.0	67.0	60.0
gradient-tight	0%	50%	0%	B	6.0	7.0	16.5	61.0	66.5	61.0

post_proc	0%	0%	50%	C	2.5	24.5	2.0	61.5	79.0	59.5
post_proc-tight	100%	100%	100%	A/B/C	0.0	0.0	0.0	66.0	63.5	61.5
model_free	0%	50%	0%	B	8.5	7.0	14.0	63.0	69.0	61.0
model_free-tight	0%	50%	0%	B	8.5	6.0	14.0	63.5	68.5	59.5
dpcc-c-tight-dt0.25	50%	50%	50%	A/B/C	0.5	0.5	1.0	75.5	75.5	75.5
dpcc-c-tight-dt0.5	100%	100%	100%	A/B/C	0.0	0.0	0.0	68.5	68.5	62.5
dpcc-c-tight-dt2.0	100%	100%	100%	A/B/C	0.0	0.0	0.0	64.5	64.5	62.0
dpcc-c-tight-dt4.0	100%	100%	100%	A/B/C	0.0	0.0	0.0	85.0	85.0	79.5

Section 4 — dt Sensitivity: Complete Data (A vs B vs C)

Full breakdown of dpcc-c-tightened dt ablation for all three models.

Top-Right-Hard

dt	A G& C;	B G& C;	C G& C;	A Steps±std	B Steps±std	C Steps±std	A Time	B Time	C Time
dt=0.25	100 %	100 %	50%	81.0±6.0	82.0±5.0	82.0±0.0	2.225	1.644	1.027
dt=0.5	0%	100 %	50%	75.5±2.5	74.5±6.5	62.0±0.0	1.668	1.163	0.884
dt=1.0 (def)	100 %	100 %	100 %	66.5±0.5	64.0±6.0	59.5±3.5	0.911	0.585	0.505
dt=2.0	100 %	100 %	50%	66.5±0.5	69.5±8.5	57.0±0.0	0.689	0.500	0.479
dt=4.0	50%	0%	0%	72.0±0.0	0.0±0.0	0.0±0.0	0.535	0.491	0.462

Top-Left-Hard

dt	A G& C;	B G& C;	C G& C;	A Steps±std	B Steps±std	C Steps±std	A Time	B Time	C Time
dt=0.25	100 %	50%	50%	70.0±3.0	89.0±7.0	70.0±0.0	1.259	1.071	1.147
dt=0.5	100 %	100 %	50%	63.5±0.5	66.5±0.5	62.0±0.0	0.937	0.563	0.811
dt=1.0 (def)	100 %	100 %	100 %	63.5±3.5	62.0±3.0	59.0±4.0	0.822	0.646	0.577
dt=2.0	100 %	100 %	50%	64.5±0.5	68.0±8.0	58.0±0.0	0.679	0.563	0.478
dt=4.0	0%	100 %	0%	107.0±2.0	120.0±6.0	0.0±0.0	0.810	0.486	0.413

Both-Hard

dt	A G& C;	B G& C;	C G& C;	A Steps±std	B Steps±std	C Steps±std	A Time	B Time	C Time
dt=0.25	50%	50%	50%	75.5±0.5	75.5±0.5	75.5±1.5	1.835	1.835	1.560
dt=0.5	100 %	100 %	100 %	68.5±4.5	68.5±4.5	62.5±4.5	1.185	1.185	1.133
dt=1.0 (def)	100 %	100 %	100 %	56.5±2.5	72.0±12.0	57.5±2.5	0.997	0.779	0.918
dt=2.0	100 %	100 %	100 %	64.5±2.5	64.5±2.5	62.0±3.0	0.737	0.737	0.739
dt=4.0	100 %	100 %	100 %	85.0±2.0	85.0±2.0	79.5±5.5	0.691	0.691	0.601

Section 5 — Summary Statistics & HP Change Assessment

5.1 Aggregate Statistics — Core Methods (13 × 3 = 39 observations)

Metric	A (Diffusion)	B (FM aw=10)	C (FM aw=1)	C-B Δ	C-A Δ
G&C; Success	0.551	0.385	0.423	+0.038	-0.128
Constraint Sat.	0.551	0.385	0.423	+0.038	-0.128
Task Success	0.987	0.885	0.872	-0.013	-0.115
Avg Violations	4.936	12.038	7.385	-4.654	+2.449
Avg Total Viol.	0.372	1.207	1.341	+0.134	+0.969
Avg Time (s/step)	0.671	0.750	0.566	-0.184	-0.104

5.2 Win Counts — G&C; Success (39 method×scenario comparisons)

A wins: 11	B wins: 5	C wins: 5	Ties/Partial: 18
------------	-----------	-----------	------------------

5.3 HP Change Assessment: action_weight 10 → 1

1. C outperforms B on both-hard — the hardest scenario

action_weight=1 achieves notably better G&C; on both-hard. C scores 50–100% on dpcc-r, dpcc-c, dpcc-t (untightened) where B scored 0–50%. The lower action weight helps the model navigate simultaneous dual-obstacle constraints more reliably.

2. C is weaker than B on top-right-hard for untightened dpcc variants

dpcc-r, dpcc-c, and dpcc-t score 0% G&C; for C on top-right-hard (identical to B's 0%). However dpcc-r-tight, dpcc-c-tight, and dpcc-t-tight all achieve 100% for C vs B's mixed 50–100%. C is more dependent on tightening for top-right.

3. C dramatically improves on top-left-hard

C achieves 100% on dpcc-c, dpcc-c-tight, dpcc-t, dpcc-t-tight on top-left-hard. B only achieved 50–100% on these. The improvement in the left-obstacle scenario is the most consistent benefit of the action_weight=1 change.

4. Baseline methods (diffuser, gradient, model_free) worsen for C on top-right

C's diffuser produces 10.3 avg total violation on top-right vs B's 7.7 — a 34% increase. gradient violations also increase. The lower action weight makes unconstrained methods less conservative on this scenario.

5. C consistently faster than B on DPCC methods

C's DPCC methods are systematically 5–20% faster per step than B (e.g. dpcc-c-tight: 0.505 vs 0.585 s/step). The action_weight change affects the SLSQP projection problem complexity.

6. dt sensitivity: C is more fragile at dt=0.25 and dt=2.0 on top-right

C achieves only 50% G&C; at dt=0.25, 0.5, 2.0 on top-right, where B achieved 100% at those settings. dt=4.0 remains 0% for both. On both-hard, C matches or improves over B across all dt values.

7. C vs A (Diffusion): C partially closes the gap

C wins 5 comparisons vs B's 5 vs A's 11. C is competitive with Diffusion on top-left-hard (matching A on dpcc-c/t variants) but Diffusion still dominates on top-right-hard across most methods.

8. Overall verdict: HP1 is a mixed improvement

action_weight=1 improves G&C; on top-left-hard and both-hard but does not help (and sometimes hurts) on top-right-hard for untightened methods. Tightened DPCC variants are consistently reliable for C. For production use, dpcc-c-tightened and dpcc-t-tightened remain the safest choices.

A = Diffusion (GaussianDiffusion/6, diffusion branch). B = FM original (flow_matching/6, action_weight=10). C = FM HP1 (flow_matching_hp_tune1/6, action_weight=1, checkpoint step 89000). All: Seed 6, avoiding-d3il, H8_K20_Dmodels.